ACADEMIC PROJECT REPORT



Ishara

Gesture-to-Text and Speech Communication App

Course Name	**Advanced Programming Laboratory**
Course Title	**CSE 2200**
Submitted by	**Md. Atif Absar**
Roll	**2307087**
Year / Semester	**2nd Year / 2nd Semester**
Department	**Computer Science and Engineering**
University	**Khulna University of Engineering & Technology (KUET)**
Version	Ishara 6.0 prototype

October 2026 · Android Studio · Java + XML

Abstract and problem statement

Abstract. Ishara is a Java-based Android accessibility prototype that helps people communicate using a small recognized gesture vocabulary, readable text, text-to-speech, and supportive communication tools. Its main interaction is to recognize a handshape through the phone camera, show the assigned demonstration meaning, combine words or phrases into sentences, and speak the completed message.

Problem statement

People who cannot speak easily, people using gestures, and people with hearing difficulties can face barriers during everyday communication. Many listeners do not understand another person's hand signals; typing can also be difficult in time-sensitive interactions. A low-cost, approachable tool is needed to support clearer two-way communication.

Objectives

- Recognize predefined hand gestures locally through an Android camera.
- Convert recognized gestures to readable Bangla demonstration phrases and spoken output.
- Combine recognized words/phrases into an editable sentence.
- Offer live captions, visual communication boards, saved phrases, and offline practice.
- Demonstrate real Java Android programming concepts across laboratory Weeks 8–11.

Core principle
Gesture Camera → Detected Handshape → Bangla Demo Meaning → Editable Sentence → Speech

Primary users

People with speech-related communication difficulties; people who prefer gesture-assisted communication; deaf or hard-of-hearing users who benefit from text captions; and family members, caregivers, and others communicating with them. Classroom use is an additional scenario, not the application's primary purpose.

02 / PROPOSED SOLUTION

Solution and application workflow

Ishara combines Android CameraX with the MediaPipe Gesture Recognizer. A pretrained model returns recognized general handshapes and hand landmarks. Java translates these categories to transparent, predefined demo phrases. Users can confirm, correct, combine, speak, save or share messages.

CameraX live video → MediaPipe on-device gesture classification

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Confidence + 21-point landmark visualization + calibration threshold

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Gesture Dictionary / correction log / context-specific demo mapping

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Java rule-based Sentence Builder → TextToSpeech / Save / Share / Peer Link

Complementary communication flow

Microphone → Android SpeechRecognizer → Live Captions / Two-way conversation → SQLite caption history → Search / Share.

Design principles

Accessible: large buttons and legible text. **Local-first:** recognition and most phrase operations happen on the device. **Understandable:** users see which handshape was recognized and can correct errors. **Lab-oriented:** the code includes explicit Activities, adapters, custom Views, SQLite operations, Services, Intents and networking examples.

> **Scope note:** This version is not a complete Bangla Sign Language translator. It recognizes seven general handshapes and assigns illustrative phrases. Actual sign-language support requires separate labeled data and evaluation with knowledgeable signers.

03 / APPLICATION CAPABILITIES

Features included in the project source

01	**Gesture Recognition Studio** CameraX preview, MediaPipe model, 7-handshape classification, confidence display, capture and Bangla speech.
02	**Gesture Calibration & Corrections** Per-class confidence controls, camera practice, human-labeled correction records in SQLite.
03	**Hand Landmark Overlay** Custom Java Canvas drawing of 21 landmarks and joint connections over live camera preview.
04	**Visual Gesture Dictionary** Searchable hand illustrations, Bangla demo meanings, tap-to-practice Camera activity.
05	**Sentence Builder & Rules** Word tiles, gesture-based sentence construction, controlled Bangla templates, speak, save and share.
06	**Live Captions & Lecture Mode** Bangla / English speech recognition, live text, saved caption sessions and export.
07	**Two-Way Conversation** Text-to-speech replies and incoming speech-to-text rendered as conversations.
08	**Quick Phrases & Picture Board** Common messages, categories, favorites, and large accessible touch targets.
09	**Login, Profiles & Adaptive Dashboard** Local registration and guest mode, account-separated storage and frequently used phrases.
10	**Gesture Learning, Accuracy & Ishara Quest** Practice lessons, confidence/success statistics, gesture quizzes and sentence puzzles.
11	**Peer Link & Real-Life Simulator** Same-Wi-Fi prototype message exchange and branching real-life communication practice.
12	**History, Settings & Lab Demonstration** SQLite CRUD, Service, BroadcastReceiver, ContentProvider, HTTP JSON and optional Firebase demonstration.

Implementation design

Programming stack

- **Java 17**: Activities, object-oriented classes, algorithms, input validation and event listeners.
- **XML**: screen structure and accessible UI components.
- **CameraX / MediaPipe**: live frame input, category labels and 21 hand landmarks.
- **TextToSpeech / SpeechRecognizer**: spoken messages and live transcripts.
- **SQLite**: message history, practice records, game progress and user corrections.

Supporting libraries

- **RecyclerView**: visual dictionary, history and chat cards.
- **Material Components**: clean UI controls and feedback.
- **Intents, Services, Receivers, ContentProvider**: Android component communication.
- **HTTP / JSON, optional Firebase**: network demonstrations.
- **Java Sockets**: same-Wi-Fi Peer Link prototype.

Local data model

Data area	Important fields / purpose
Messages	Text, type, timestamp; saved communication and history.
Captions	Session title, transcript body, language, timestamp.
Gesture practice	Expected class, detected class, confidence, outcome, timestamp.
Gesture corrections	Predicted class, corrected class, confidence, timestamp, separate database.
Phrase usage	Phrase, category, usage count; adaptive shortcuts.
Profiles / games	Local login data, quest scores, scenario progress.

Sensitive conversation records remain on-device by default. Optional network and Firebase modules require extra setup. Local accounts are not remote authenticated cloud accounts.

05 / LAB ALIGNMENT

Advanced Programming Laboratory coverage

Week	Topics from syllabus	Evidence in Ishara
8	Android Studio architecture, Manifest, lifecycle, XML, ViewBinding, listeners, explicit and implicit intents.	Welcome, Gesture, Dictionary and Calibration Activities; permission flows; buttons; share text.
9	RecyclerView, adapters, ViewHolders, custom views, mobile UX.	DictionaryAdapter, history/chat lists, Gesture Landmark Canvas overlay, confidence visualization.
10	Services, BroadcastReceivers, ContentProviders, permissions.	SpeechService, battery receiver, message provider, microphone and camera permissions.
11	Networking, HTTP, JSON, SQLite, background operations, advanced APIs, Firebase.	JSON client, phrase data, SQLite corrections and history, model downloads, MediaPipe, optional Firebase and Peer Link.

Java programming evidence

- Encapsulation, inheritance and custom classes, including `GestureCalibration`, `SentenceRules`, `DictionaryAdapter`, and SQLite helpers.
- Collections (`List`, `Map`, `ArrayList`), String transformations, conditional rules and search filtering.
- Multi-threaded frame processing, callback-based event handling, exception handling and object lifecycle management.
- Clear separation between Android UI code, reusable logic, persistence, services and networking.

Why Java remains central: Ishara does not require a separate C++ module. The project's Android APIs, camera callback logic, database operations and app interactions are implemented through Java and XML.

06 / ILLUSTRATED DEMO REFERENCE

Seven supported handshapes & meanings

These illustrations show the **general handshape categories** detected by the stock MediaPipe model, alongside the meanings assigned by Ishara's Everyday demo context.



Thumbs Up

হ্যাঁ

Yes / agreement

Thumb_Up



Thumbs Down

না

No / disagreement

Thumb_Down



Open Palm

একটু থামুন

Please stop / pause

Open_Palm



Closed Fist

আমার সাহায্য দরকার

I need help

Closed_Fist



Victory

ধন্যবাদ

Thank you

Victory



Pointing Up

একটু অপেক্ষা করুন

Please wait

Pointing_Up



I-Love-You handshape

আপনাকে ভালোবাসি

I love you

ILoveYou

> **Important:** The illustrations are simplified visual references, not a validated Bangla Sign Language lesson. The same gesture may map to different demo phrases in different app contexts. Recognition quality depends on lighting, camera view, hand positioning and model limitations.

Testing, limitations and conclusion

Testing plan

- Build the Android Studio project with SDK 35 and Gradle JDK 21; verify installation and all navigation routes.
- Use a physical Android camera to evaluate all seven gestures in multiple lighting conditions. Record confidence, false positives and corrections.
- Build the message “আমি পানি চাই” in the Sentence Builder, apply the rule suggestion, save and speak it.
- Test dictionary search, adjustable thresholds, landmark overlay toggle and database persistence after restart.
- Test microphone permissions and Live Captions in Bangla and English with an installed speech recognition service.
- Test login, local data isolation, saved history, Quest and optional Peer Link on two devices.

Limitations

The stock model recognizes only seven general handshapes. Calibrating thresholds does not retrain the neural network. The 21-point overlay is an approximate visual aid that requires physical-device alignment testing. Continuous captioning and Bangla speech output depend on device services. Peer Link is an unencrypted local demonstration. The project source passes static checks, but a full Android Gradle build and usability evaluation have not been completed in this report environment.

Conclusion

Ishara demonstrates how Java-based Android programming can support practical, more inclusive communication. Its main contribution is a transparent gesture-to-message workflow supported by sentence composition, speech output, visual learning and calibration. Its secondary modules broaden accessibility while providing demonstrable examples of the full laboratory syllabus.

References

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Prepared for Advanced Programming Laboratory (CSE 2200), Department of Computer Science and Engineering, Khulna University of Engineering & Technology.